

3. The picture shows a smoke detector on the ceiling of a room.
It contains the radioactive isotope americium-241 (Am-241) which emits alpha particles.



Smoke getting into the detector absorbs alpha particles.
This changes the electric current inside the detector and the alarm sounds.

- (a) Underline the phrase that correctly completes the sentence.

An alpha particle is a (**helium nucleus / hydrogen atom / fast moving electron**). [1]

- (b) Explain why the americium-241 inside the detector is not a risk to humans when the detector is used on the ceiling. [2]

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- (c) Explain why the smoke detector would not work if the radioactive isotope only emitted gamma rays. [2]

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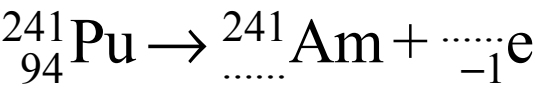
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- (d) Am-241 is produced when plutonium-241 (Pu-241) decays.
- (i) Select values from the box to answer the questions that follow.

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I. **Complete** the decay equation below. [2]



II. State the number of **protons** in the nucleus of a Pu-241 atom. [1]

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III. Calculate the number of **neutrons** in the nucleus of a Pu-241 atom. [1]

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(ii) Circle the correct name for the other particle that is produced when Pu-241 decays. [1]

(gamma particle / beta particle / helium particle)

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